

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code:	CSA10 / CSA1008	Course Title:	Software Engineering
CO Assessed:	CO2	Assessment:	Assessment Tool 2 – Scenario-Based Requirement Analysis
Weightage:	20%	Total Marks:	25
CO2 Statement:	<i>Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills</i>		
Student Name:	K.Bhavana	Reg. No.:	192524178
Date:	1-08-2026	Duration:	60 minutes

Code of Conduct

I K. Bhavana (192524178) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

K. Bhavana

Annexure A – Scenario-Based Requirement Analysis

Instructions to Students:

Each student will be assigned an individual software project problem statement. Analyze the given scenario and prepare a complete Software Requirement Analysis document by following the steps below.

Question

Based on the **assigned problem statement**, perform a **Scenario-Based Requirement Analysis** and prepare a Software Requirement Specification (SRS) document. Your analysis should include the following:

- 1. Study and Analyze the Scenario**
 - Carefully understand the given problem statement.
 - Identify the business domain, stakeholders, existing challenges, and system objectives.
- 2. Identify Functional and Non-Functional Requirements**
 - List all functional requirements required by the proposed system.
 - Identify suitable non-functional requirements such as performance, security, reliability, usability, scalability, maintainability, and availability.
- 3. Identify Actors and Use Cases**
 - Identify all primary and secondary actors interacting with the system.
 - List the major use cases associated with each actor.
 - Draw a Use Case Diagram representing the interactions between actors and the system.
- 4. Prepare the Software Requirement Specification (SRS)**

Develop an SRS document containing:

 - Introduction

- Problem Description
- Scope of the System
- Functional Requirements
- Non-Functional Requirements
- Use Case Descriptions
- Data Requirements
- External Interface Requirements
- System Features

5. Identify Assumptions and Constraints

- List the assumptions considered while developing the system.
- Identify technical, operational, economic, legal, and time-related constraints affecting the project.

6. Validate Requirement Completeness and Consistency

- Verify whether all stakeholder requirements have been addressed.
- Check for ambiguity, redundancy, inconsistency, and missing requirements.
- Suggest improvements to enhance the quality and completeness of the requirement specification.

1. Introduction

1.1.Purpose

This Software Requirement Specification (SRS) document defines the functional and non-functional requirements for the Women Entrepreneurship Support Portal (WESP). It is intended to guide the design, development, and validation of a system that connects women entrepreneurs with government funding schemes, skill-development training, and business mentorship through a single digital platform.

1.2.Intended Audience

This document is intended for the development team, project stakeholders (government department officials and administrators), quality assurance personnel, and academic evaluators assessing this scenario-based requirement analysis.

1.3.Business Domain

The system operates in the entrepreneurship-support and public-welfare domain, specifically women-focused economic empowerment. It supports government departments in publishing and disbursing funding schemes, delivers structured training and mentorship to women entrepreneurs, and gives administrators oversight of users, schemes, and outcomes across the platform.

2. Problem Description

Women entrepreneurs currently discover government funding schemes, training programs, and mentorship opportunities through fragmented, largely offline channels: scheme information is scattered across department notices, funding applications are tracked manually, and there is no structured way to connect an entrepreneur with a suitable business mentor. This leads to low scheme awareness, delayed or lost applications, inconsistent verification of eligibility, and limited access to guidance — particularly for entrepreneurs in areas with limited access to government offices.

The proposed Women Entrepreneurship Support Portal addresses these challenges by providing a centralized digital platform where entrepreneurs can register, apply for and track funding, enroll in training, and connect with mentors; where government officers can publish schemes and verify and approve applications; where mentors can manage mentorship requests and guidance sessions; and where administrators can manage users, schemes, training programs, and reporting — reducing manual effort and improving transparency and reach.

3. Scope of the System

The system covers entrepreneur self-registration, login, and profile management; funding-application submission and tracking; browsing of government schemes; enrollment in and viewing of training programs; mentor discovery, requesting, and session scheduling; automated notifications; funding approval and scheme publishing by

government officers; and user, mentor, scheme, and training-program management with report generation for administrators.

Out of scope for this phase: direct disbursement of funds through the portal (approved funding is disbursed via existing government payment channels), integration with external banking/KYC verification systems beyond basic document upload, and a full learning-management system for training content (the portal manages enrollment and scheduling, not course delivery).

4. Stakeholders

The primary stakeholders of the system, drawn from the Use Case model, are:

Actor	Type	Description
Women Entrepreneur	Primary	A woman running or starting a business who registers, applies for funding, enrolls in training, and connects with mentors.
Government Officer	Primary	Department official who publishes schemes, verifies eligibility, and approves or rejects funding applications.
Mentor	Primary	Experienced business professional who accepts mentorship requests, provides guidance, and schedules sessions with entrepreneurs.
Administrator	Primary	Staff who manage platform users, mentors, training programs, government schemes, and funding opportunities, and generate reports.
Notification Service	Secondary (external system)	External SMS/email/push gateway invoked by the system to deliver alerts to entrepreneurs and other users.

5. Functional Requirements

The following functional requirements were derived from the scenario and are reflected in the system's use-case and class model (see attached UML artifacts).

ID	Requirement	Description	Priority
FR-01	Entrepreneur Registration	The system shall allow a woman entrepreneur to register by providing name, business name, email, phone number, and address, and generate a unique Entrepreneur ID.	High
FR-02	User Login / Logout	The system shall authenticate Entrepreneurs, Government Officers, Mentors, and Administrators using valid credentials and allow them to securely log out.	High

ID	Requirement	Description	Priority
----	-------------	-------------	----------

FR-03	Update Profile	The system shall allow an entrepreneur to view and update their profile and business details after registration.	Medium
FR-04	Apply for Funding	The system shall allow a registered entrepreneur to apply for an available government funding scheme by submitting the required details.	High
FR-05	Track Funding Application	The system shall allow an entrepreneur to view the current status (Submitted, Under Review, Approved, Rejected) of their funding application.	High
FR-06	View Government Schemes	The system shall allow entrepreneurs and government officers to browse published government schemes and their eligibility criteria.	High
FR-07	Enroll in Training	The system shall allow an entrepreneur to enroll in an available skill-development training program.	Medium
FR-08	View Training Programs	The system shall allow an entrepreneur to view available and enrolled training programs and their schedules.	Medium
FR-09	Request / Connect with Mentor	The system shall allow an entrepreneur to browse mentors and send a mentorship request.	High
FR-10	Receive Notifications	The system shall automatically send application-status, training, and mentorship notifications to entrepreneurs via SMS/email/push.	Medium
FR-11	Manage Mentorship Requests	The system shall allow a mentor to view incoming mentorship requests and accept or decline them.	High
FR-12	Provide Guidance & Schedule Sessions	The system shall allow a mentor to provide business guidance and schedule mentoring sessions with accepted entrepreneurs.	Medium
FR-13	Approve Funding	The system shall allow a government officer to verify and approve or reject a submitted funding application.	High
FR-14	Publish Government Schemes	The system shall allow a government officer to create and publish new funding schemes with eligibility criteria and validity period.	High
FR-15	Manage Training Programs	The system shall allow a government officer or administrator to create, update, and manage training program listings.	Medium
FR-16	Manage Users	The system shall allow an administrator to create, update, deactivate, and assign roles to system users.	Medium
FR-17	Manage Mentors	The system shall allow an administrator to onboard, verify, update, and remove mentor accounts.	Medium
FR-18	Manage Schemes & Funding Opportunities	The system shall allow an administrator to oversee and, where needed, edit or archive government schemes and funding opportunities.	Medium
FR-19	Generate Reports	The system shall allow an administrator to generate, view, and export reports on registrations, funding approvals, training enrollment, and mentorship activity.	Medium

6. Non-Functional Requirements

Category	Requirement
Performance	The system shall respond to funding-application submission and status-update requests within 3 seconds under normal load (up to 500 concurrent users).
Security	The system shall encrypt sensitive entrepreneur personal and financial data in transit (TLS) and at rest, enforce role-based access control across the four user roles, and log all administrative and funding-approval actions.
Reliability	The system shall maintain 99.5% uptime during working hours and ensure no loss of funding or training records due to system failure through regular backups.
Usability	The system shall provide a simple, mobile-friendly interface accessible to entrepreneurs with basic digital literacy, in at least one regional language and English.
Scalability	The system shall support horizontal scaling of the database and server tier to accommodate growth from a pilot district to a state-wide or national rollout.
Maintainability	The system shall use a modular architecture (UI, Authentication, Funding, Training, Mentorship, Notification, Reporting, Database) so modules can be updated independently.
Availability	The system shall be available 24/7 for entrepreneur self-service functions (registration, applications, browsing schemes/training), with scheduled maintenance windows communicated in advance.
Interoperability	The system shall expose APIs / export formats compatible with government scheme-management and MIS reporting systems for data exchange.
Data Integrity	The system shall validate all funding applications and mentorship requests (eligibility checks, duplicate-request prevention) before saving to prevent inconsistent or duplicate entries.

7. Actors and Use Cases

7.1. Actors

Primary actors (Women Entrepreneur, Government Officer, Mentor, Administrator) interact directly and frequently with the system to perform core tasks. The Notification Service is a secondary external system invoked by the platform rather than a human actor.

7.2. Use Case Summary

Use Case	Actor(s)	Description
Register	Women Entrepreneur	Allows a new entrepreneur to create an account with personal and business details.
Login / Logout	All actors	Authenticates a user and terminates an active session securely.
Update Profile	Women Entrepreneur	Allows an entrepreneur to view and edit their profile and business details.
Apply for Funding	Women Entrepreneur	Allows an entrepreneur to submit an application for a government funding scheme.

Use Case	Actor(s)	Description
----------	----------	-------------

Track Funding Application	Women Entrepreneur	Displays the current status of a submitted funding application.
View Government Schemes	Women Entrepreneur, Government Officer	Displays published funding schemes and eligibility criteria.
Enroll in Training	Women Entrepreneur	Allows an entrepreneur to enroll in an available training program.
View Training Programs	Women Entrepreneur	Displays available and enrolled training programs and schedules.
Connect with Mentor / Request Mentor	Women Entrepreneur	Allows an entrepreneur to browse mentors and send a mentorship request.
Receive Notifications	Women Entrepreneur	Sends automated alerts for application status, training, and mentorship updates.
View / Accept Mentorship Requests	Mentor	Allows a mentor to review incoming requests and accept or decline them.
Provide Business Guidance / Schedule Session	Mentor	Allows a mentor to guide an entrepreneur and schedule mentoring sessions.
Approve Funding	Government Officer	Verifies and approves or rejects a submitted funding application.
Publish Government Schemes	Government Officer	Creates and publishes new funding schemes.
Manage Training Programs	Government Officer, Administrator	Creates, updates, and manages training program listings.
Manage Users	Administrator	Creates, updates, deactivates, and assigns roles to system users.
Manage Mentors	Administrator	Onboards, verifies, updates, and removes mentor accounts.
Manage Schemes & Funding Opportunities	Administrator	Oversees, edits, or archives government schemes and funding opportunities.
Generate Reports	Administrator	Produces registration, funding, training, and mentorship activity reports.

7.3. Use Case Diagrams

To keep each diagram readable, the use-case diagram is split into four views — one per actor — rather than a single combined diagram. Use cases shared across actors (e.g., Login) appear in more than one view.

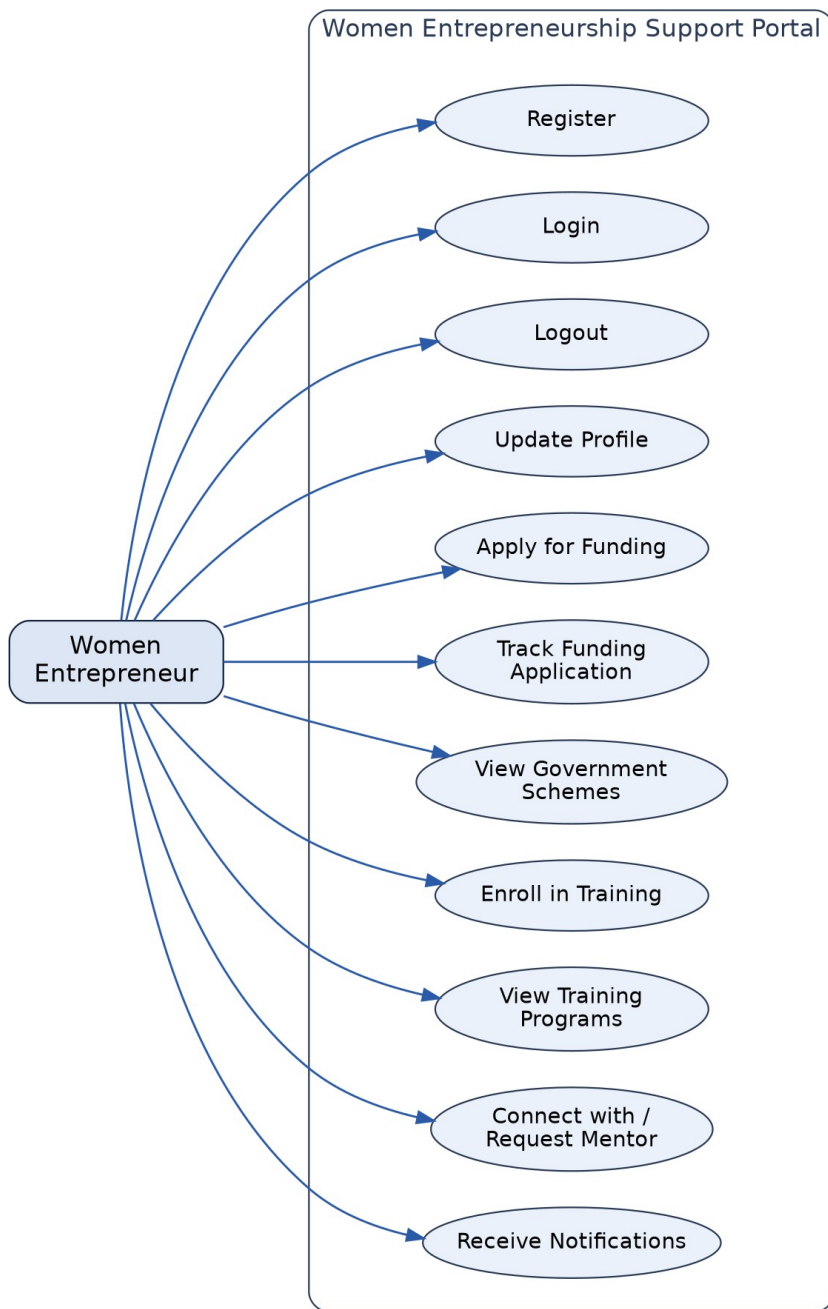


Figure 1a: Use Case Diagram — Women Entrepreneur

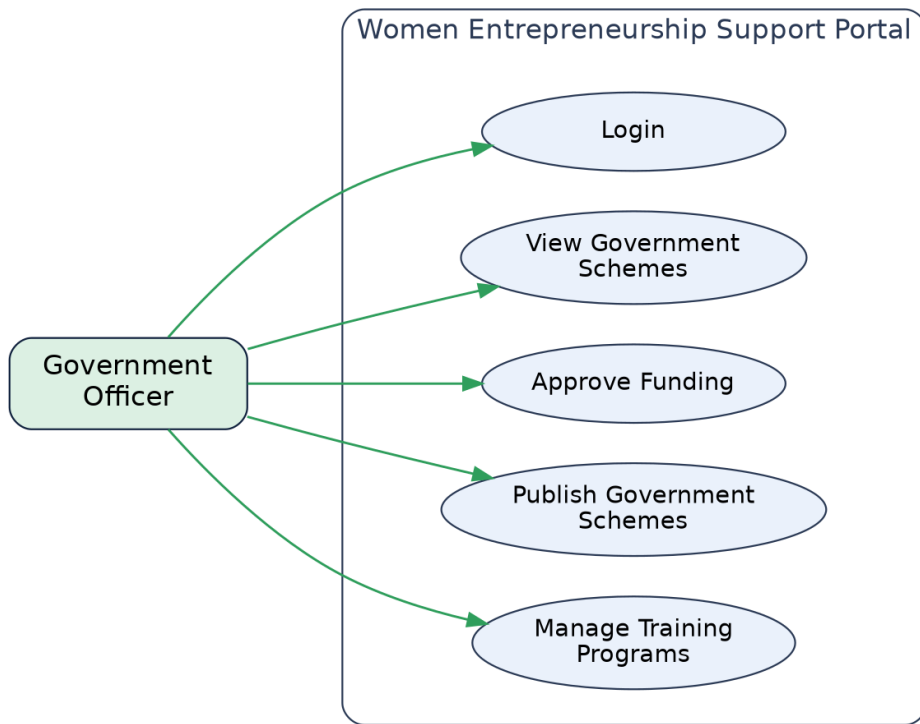


Figure 1b: Use Case Diagram — Government Officer

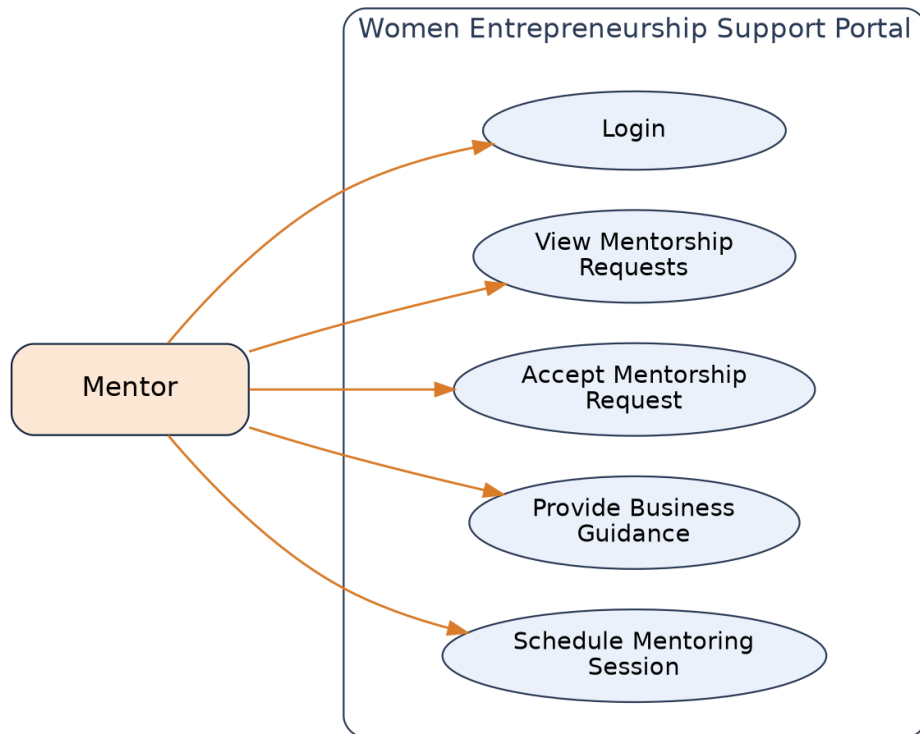


Figure 1c: Use Case Diagram — Mentor

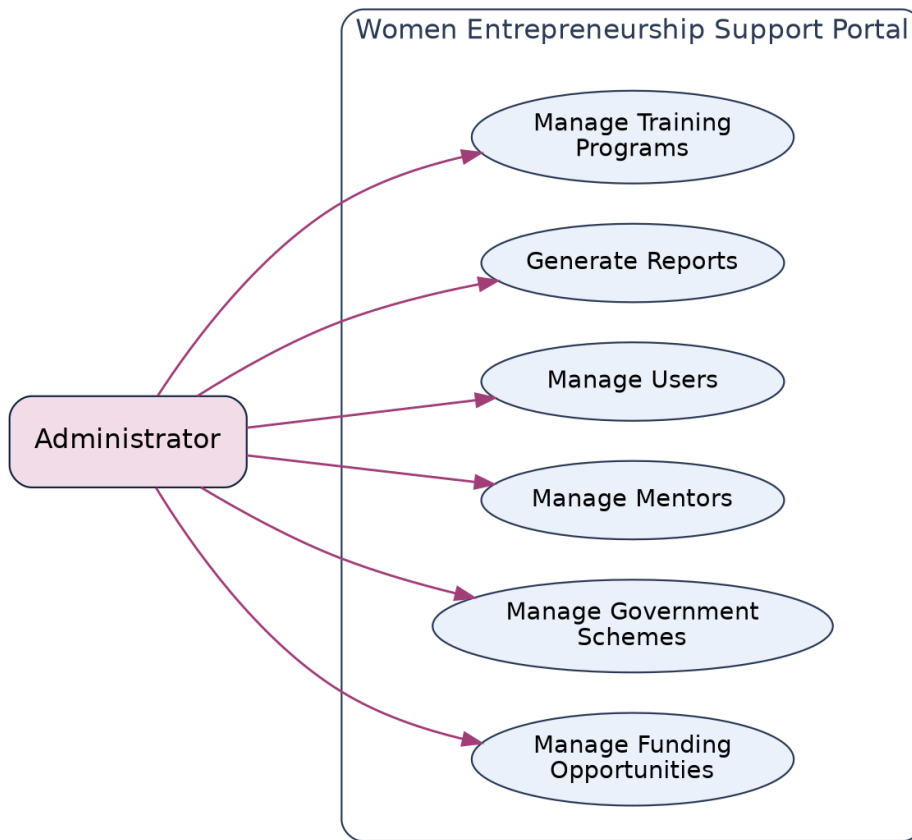


Figure 1d: Use Case Diagram — Administrator

7.4.Detailed Use Case Descriptions

UC-01: Register

Actor: Women Entrepreneur

Preconditions: The entrepreneur has access to the portal and is not already registered.

Main Flow:

1. Entrepreneur opens the registration page.
2. Entrepreneur enters name, business name, email, phone number, and address.
3. System validates the input data (mandatory fields, email/phone format).
4. System creates a unique Entrepreneur ID and stores the record in the database.
5. System displays a registration confirmation to the entrepreneur.

Alternate/Exception Flow: If validation fails (e.g., duplicate email, missing field), the system displays an error message and requests correction.

Postconditions: A new entrepreneur record exists in the system and the entrepreneur can log in.

UC-02: Apply for Funding

Actor: Women Entrepreneur

Preconditions: The entrepreneur is registered and logged in.

Main Flow:

1. Entrepreneur selects “Apply for Funding” and chooses an available scheme.
2. System displays the scheme’s eligibility criteria and required details.
3. Entrepreneur submits the funding application with supporting details.
4. System validates the submission and records the application with status “Submitted”.
5. System notifies the entrepreneur that the application has been received.

Alternate/Exception Flow: If the entrepreneur does not meet stated eligibility criteria, the system flags the mismatch before final submission.

Postconditions: A funding-application record is created and linked to the entrepreneur, ready for government-officer review.

UC-03: Approve Funding

Actor: Government Officer

Preconditions: The officer is logged in and a funding application is pending review.

Main Flow:

1. Officer opens the list of pending funding applications.
2. Officer reviews applicant details against scheme eligibility criteria.
3. Officer approves or rejects the application, optionally adding remarks.
4. System updates the application status and notifies the entrepreneur.

Alternate/Exception Flow: If required supporting information is missing, the officer marks the application as “Needs Information” and the system prompts the entrepreneur to resubmit.

Postconditions: The funding application’s status is updated and the entrepreneur is informed of the outcome.

UC-04: Request and Accept Mentorship

Actor: Women Entrepreneur, Mentor

Preconditions: The entrepreneur is registered and logged in; at least one mentor is available on the platform.

Main Flow:

1. Entrepreneur browses available mentors and sends a mentorship request.
2. System notifies the selected mentor of the new request.
3. Mentor reviews the request and accepts or declines it.
4. If accepted, system links the mentor and entrepreneur and enables session scheduling.
5. Mentor schedules a mentoring session and provides business guidance.

Alternate/Exception Flow: If the mentor declines, the system notifies the entrepreneur and suggests alternative available mentors.

Postconditions: A mentor-entrepreneur relationship is established and, where accepted, a mentoring session is scheduled.

UC-05: Generate Reports

Actor: Administrator

Preconditions: The administrator is logged in with report-generation privileges.

Main Flow:

1. Administrator selects report type (registrations, funding, training, mentorship) and date range.
2. System compiles data from the relevant records.
3. System displays the report and offers export options.

Alternate/Exception Flow: If no data exists for the selected range, the system informs the administrator rather than generating an empty report.

Postconditions: A report is generated and made available for viewing or export.

8. Data Requirements

The system's core data entities, derived from the domain class model, are summarized below. In the current UML model, WomenEntrepreneur and GovernmentOfficer are fully specified with attributes and operations; FundingApplication, GovernmentScheme, TrainingProgram, Mentor, Administrator, and Notification currently exist in the model as stub classes with no attributes defined — the attributes shown for those entities below are recommended additions needed before implementation (see Section 12).

Entity	Key Attributes	Notes
WomenEntrepreneur	entrepreneurID (PK), name, businessname, email, phone, address	Modeled with full attributes and operations (register, login, applyfunding, enrolltraining, requestMentor, updateProfile, logout).
GovernmentOfficer	officerID (PK), name, department, email	Modeled with full attributes and operations (verifyRegistration, verifyApplication, approveFunding, publishScheme, generateReport).
FundingApplication *	applicationID (PK), entrepreneurID (FK), schemeID (FK), amountRequested, status, submittedDate, decisionRemarks	Recommended — stub class in current model; status should be one of Submitted, Under Review, Approved, Rejected.
GovernmentScheme *	schemeID (PK), schemeName, eligibilityCriteria, fundingAmount, validFrom, validTo, publishedBy (FK)	Recommended — stub class in current model.
TrainingProgram *	programID (PK), title, description, startDate, endDate, capacity, managedBy (FK)	Recommended — stub class in current model.
Mentor *	mentorID (PK), name, expertise, email, phone, availabilityStatus	Recommended — stub class in current model.
Administrator *	adminID (PK), name, email	Recommended — stub class in current model.
Notification *	notificationID (PK), recipientID (FK), message, channel, sentDate, status	Recommended — stub class in current model; channel should be one of SMS, Email, Push.

9. External Interface Requirements

Interface Type	Description
User Interface	Web portal and mobile-responsive interface for entrepreneurs; desktop/tablet interface for government officers, mentors, and administrators. Supports English and at least one regional language.
Hardware Interface	Standard user devices (smartphone/PC/tablet) and a central database server; no specialized hardware required.
Software Interface	Integrates with an SMS/email gateway API for notifications; exports reports in PDF format; may interface with a government scheme-management or MIS system.
Communication Interface	HTTPS for all client-server communication; RESTful APIs between the User Interface, Funding, Training, Mentorship, Reporting, and Notification modules and the central Database.

10. System Features

Feature	Description
Entrepreneur Self-Service Portal	Enables entrepreneurs to register, log in, update their profile, apply for funding, track applications, enroll in training, and connect with mentors.
Funding Application & Approval	Provides government officers a streamlined workflow to review, approve, or reject funding applications against scheme eligibility.
Training & Scheme Management	Allows government officers and administrators to publish schemes and manage training-program listings.
Mentorship Matching	Lets entrepreneurs request mentors and mentors accept requests, provide guidance, and schedule sessions.
Reporting & Analytics	Generates registration, funding, training, and mentorship reports for administrators, exportable as PDF.
Notification Service	Sends automated SMS/email/push alerts for application status, training, and mentorship updates.

11. Assumptions and Constraints

11.1.Assumptions

- Entrepreneurs have access to a smartphone, computer, or assistance from a local support center to use the system.
- Government departments provide accurate and timely scheme information for publishing on the platform.
- Mentors volunteering or contracted through the program are verified before onboarding.
- Users provide valid and truthful personal and business information during registration.
- The SMS/email gateway used for notifications is reliable and has acceptable delivery latency.

11.2.Constraints

Type	Constraint
Technical	The system must run on commonly available browsers/devices without requiring specialized equipment.
Operational	The system must be usable by entrepreneurs and government officers with limited prior IT training, requiring minimal onboarding.
Economic	Development and hosting costs must remain within the program’s allocated IT budget; open-source or low-cost technologies are preferred.
Legal / Regulatory	The system must comply with applicable data-privacy regulations and government data-protection guidelines for storing entrepreneur personal and financial information.
Time	The system must be deployed before the start of the next scheduled funding-scheme enrollment cycle.

12. Requirement Validation — Completeness and Consistency

The requirement set and the accompanying UML model were reviewed against the scenario to check for completeness, consistency, ambiguity, and redundancy. Key findings are summarized below.

Check	Observation	Status
Completeness	All actors identified in the use-case model (Women Entrepreneur, Government Officer, Mentor, Administrator) have at least one corresponding functional requirement.	Addressed
Consistency	Terminology (e.g., “Funding Application”, “Mentorship Request”) is used uniformly across FR, NFR, use cases, and data requirements.	Addressed
Model gap	FundingApplication, GovernmentScheme, TrainingProgram, Mentor, Administrator, and Notification are defined only as bare class boxes in the UML model, with no attributes or operations — unlike WomenEntrepreneur and GovernmentOfficer, which are fully modeled.	Open — needs completion
Association anomaly	The model links Government Officer to “Apply for Funding”, but the officer’s own operations (verifyApplication, approveFunding) suggest the officer should instead be linked to a “Verify/Approve Funding” use case rather than applying for funding.	Needs review
Ambiguity	“Connect with Mentor” and “Request Mentor” appear as two separate use cases for the entrepreneur with overlapping intent; this SRS treats them as a single flow (UC-04) but the model should clarify whether they are distinct steps or the same action.	Needs review
Redundancy	Login/Logout was consolidated into a single functional requirement (FR-02) instead of being duplicated per actor.	Resolved
Missing requirements	An audit-logging non-functional requirement (Security) was added after review to support accountability for funding-approval actions.	Added
Testability	Each functional requirement is stated with a clear, observable system action so it can be mapped to a test case.	Addressed

12.1.Suggested Improvements

- Complete the class model: add attributes and operations for FundingApplication, GovernmentScheme, TrainingProgram, Mentor, Administrator, and Notification (see recommended attributes in Section 8).
- Resolve the Government Officer → “Apply for Funding” association — replace or supplement it with a “Verify Application” / “Approve Funding” relationship that matches the officer’s modeled operations.
- Clarify whether “Connect with Mentor” and “Request Mentor” are one use case or two sequential steps, and update the use-case diagram accordingly.
- Define specific performance benchmarks (e.g., exact concurrent-user targets) through a load-testing exercise before final sign-off.
- Add explicit non-functional requirements for accessibility (WCAG compliance) to support entrepreneurs with disabilities.

13. Conclusion

This SRS document captures the functional and non-functional requirements, actors, use cases, data entities, and constraints for the Women Entrepreneurship Support Portal, based on scenario-based requirement analysis of the accompanying UML model. The requirements are traceable to the use-case and class diagrams, and the validation in Section 12 identifies specific gaps — chiefly the underspecified stub classes and one questionable actor-use-case association — that should be resolved before this baseline is used for system design.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modeling	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification	SRS is well-structured, complete, professionally	SRS is organized with minor missing sections or formatting	SRS includes only essential	SRS is incomplete, poorly organized,

		issues.	sections with limited detail.	or
--	--	---------	-------------------------------------	----